WCS 129: Upper respiratory tract infections (URI or URTI)

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Learning Objectives

- Define URTIs
- Be familiar with one of the commonest health problems
- Problem solving of common respiratory symptoms
- Diagnose URIs
- Evidence-based management of URTIs

What are URTIs?

"self-limited irritation and swelling of the upper airways with associated cough and no signs of pneumonia, in a patient with no other condition that would account for their symptoms, or with no history of chronic obstructive pulmonary disease, emphysema, or chronic bronchitis"

"involve the nose, sinuses, pharynx, larynx, and the large airways."

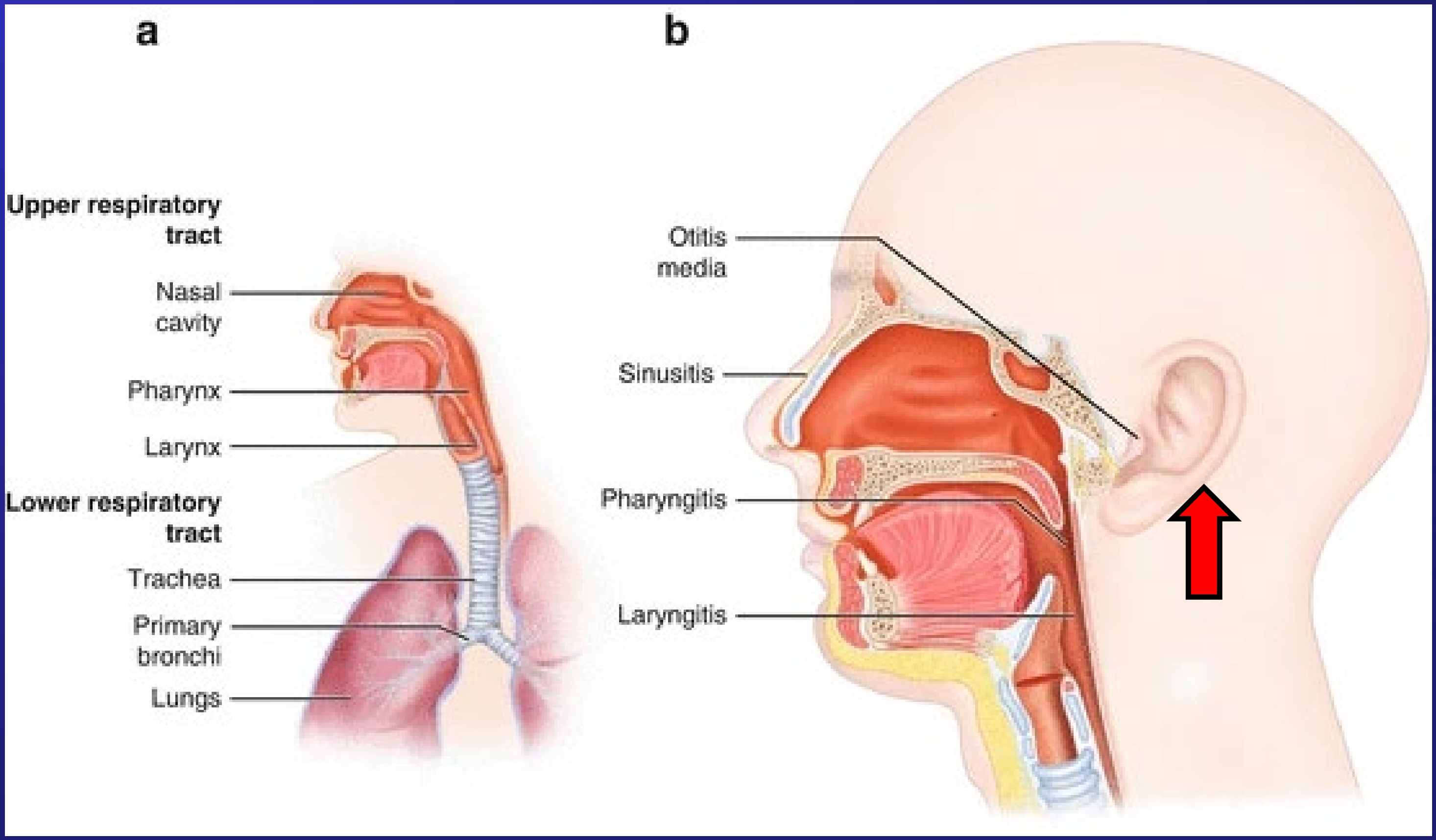
*Reference:
Thomas M, Bomar PA. Upper Respiratory Tract Infection. [Updated 2023 Jun 26]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK532961/

What are URTIs?

"involve the mucous membranes lining the upper respiratory tract from the nostrils and the mouth to the vocal cords in the larynx, also including the paranasal sinuses <u>and the middle ear</u>"

*Reference:
Calderaro A, Buttrini M, Farina B, Montecchini S, De Conto F, Chezzi C. Respiratory Tract Infections and Laboratory Diagnostic Methods: A Review with A Focus on Syndromic Panel-Based Assays. *Microorganisms*. 2022; 10(9):1856. https://doi.org/10.3390/microorganisms10091856

The Upper Respiratory Tract



*Reference:
Guibas, G.V., Papadopoulos, N.G. (2017). Viral Upper Respiratory Tract Infections. In: Green, R. (eds) Viral Infections in Children, Volume II. Springer, Cham. https://doi.org/10.1007/978-3-319-54093-1_

What are URTIs?

- Caused by variety of viruses / bacteria
- Results in various diseases such as:
  - Common cold (ex rhinovirus, RSV, adenovirus)
  - Influenza (influenza)
  - Acute bronchitis
  - Pharyngitis (ex GAS)

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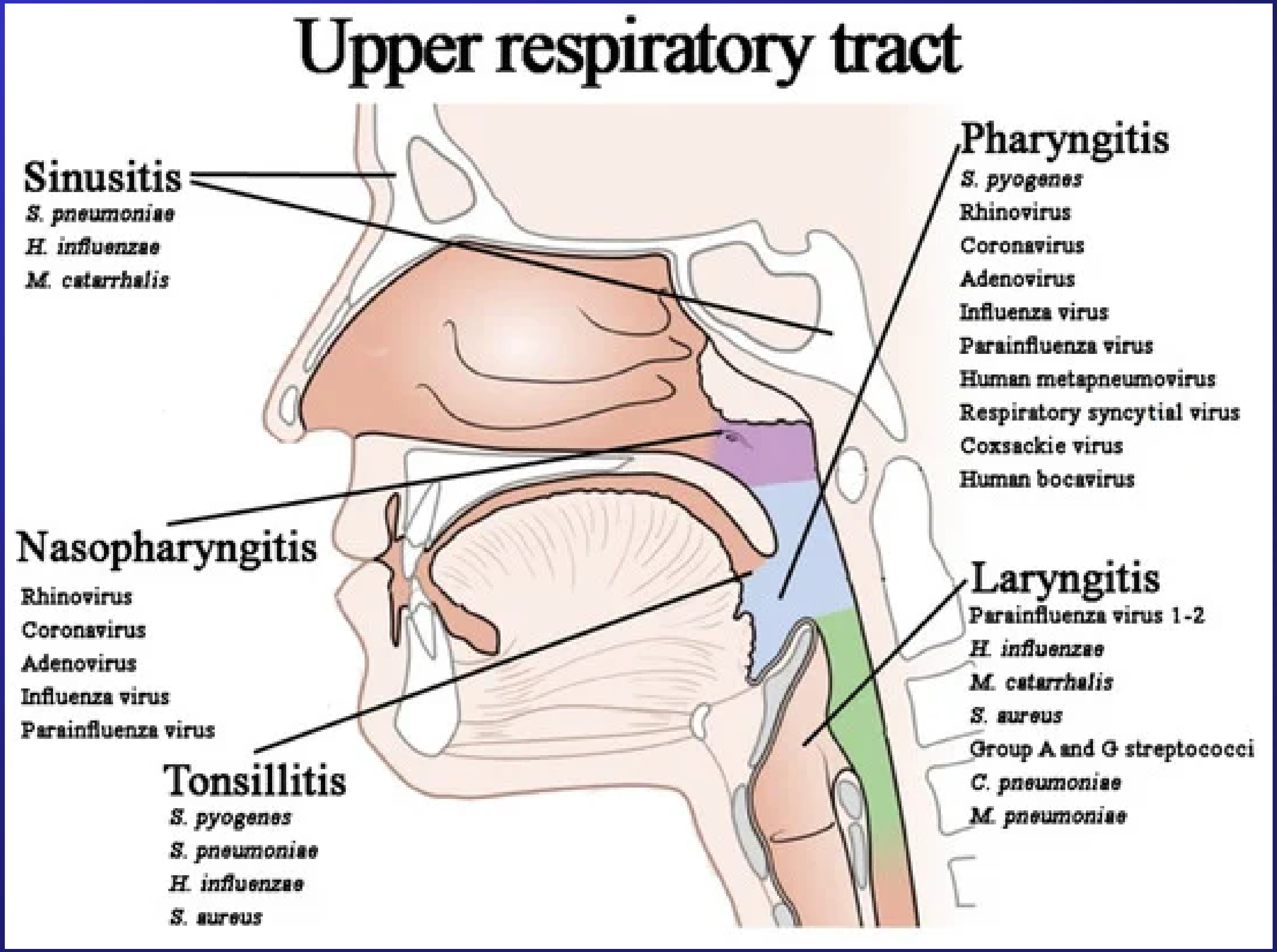

Reproduced from Calderaro A, Buttrini M, Farina B, Montecchini S, De Conto F, Chezzi C. Respiratory Tract Infections and Laboratory Diagnostic Methods: A Review with A Focus on Syndromic Panel-Based Assays. *Microorganisms*. 2022; 10(9):1856. https://doi.org/10.3390/microorganisms10091856

Upper Respiratory Tract Infections

- **>90% viral**
  - Rhinovirus
  - Adenovirus
  - Influenza
  - Parainfluenza
  - Respiratory syncytial
  - Coronavirus
- **<10% bacterial**
  - Gp A Beta Haemolytic Streptococcus
  - Haemophilus influenzae
  - Strep. pneumonia
  - Branhamella catarrhalis
  - Staphylococcus aureus

Learning Objectives

- **Be familiar with one of the commonest health problems**
- **Problem solving of common respiratory symptoms**
- **Diagnose URIs**
- **Evidence-based management of URTIs**

Disease Burden of URIs

- The most common infection in human
- Annual incidence: 2-3x/yr (adult), up to 8x/yr (kids) (source - stats pearl)
- Rarely lethal
- Often self-limiting and short in duration
- Significant health & economic burden
  - *Doctor consultations (~900,000 annual attendance in public primary care clinics*)*
  - *Manpower loss, missed days of school*
- Reason for consultation often: relief of symptoms

*Reference:
Thomas M, Bomar PA. Upper Respiratory Tract Infection. [Updated 2023 Jun 26]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK532961/

URTIs are Common

"Common problems come first"

Primary Care Problems by System

HK primary care morbidity survey 2007-08, Lo YYC et al. *HK Pract* 2010

- Respiratory 36.2%
- Cardiovascular 12.8%
- Digestive 9.8%
- Endocrine 9.3%
- Skin 7.4%
- Musculoskeletal 7.1%
- General 4.5%
- Female / male genital 2.9%
- Psychological 2.6%
- Eye 2.0%
- Neurological 1.6%
- Urological 1.4%
- Ear 1.1%
- Blood 0.5%
- Reproductive 0.5%
- Social problems 0.2%

Top Ten health problems

HK Survey 2007-08 (57.0%)

1. URI (26.4%)
2. Hypertension (10.0%)
3. DM (4.0%)
4. Gastroenteritis (3.9%)
5. Lipid disorders (2.7%)
6. Dermatitis (2.7%)
7. Acute Bronchitis (2.4%)
8. Immunization (1.9%)
9. Physical check-up (1.7%)
10. Allergic rhinitis (1.3%)

Public GOPC (64.3%)

1. Hypertension (25.7%)
2. URI (12.0%)
3. DM (10.4%)
4. Lipid disorders (5.3%)
5. Dermatitis (2.3%)
6. Old CVA (2.3%)
7. OA Knees (1.7%)
8. Dyspepsia (1.6%)
9. Gout (1.5%)
10. IHD(1.5%)

Learning Objectives

- Be familiar with one of the commonest health problems
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How do URTIs present?

- Cough
- Sore throat
- Runny nose
- Nasal congestion
- Headache
- Low-grade fever
- Facial pressure
- Sneezing
- Malaise
- Myalgias

usually begins 1-3 days after exposure

lasts 7–10 days, can persist up to 3 weeks.

BUT: Not all cough, runny nose & sore throat are URTIs.

*Reference:
Thomas M, Bomar PA. Upper Respiratory Tract Infection. [Updated 2023 Jun 26]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK532961/

URIs- Symptoms for Diagnosis

Respiratory

- Cough/Sputum
- Runny nose/ sneeze
- Nasal congestion
- Sore throat
- Hoarseness
- Nasal congestion
- Sneezing

Infection

- Chills
- General malaise
- Fever
- Loss of appetite
- Nausea/ vomiting
- (Dyspepsia/ abd pain)

Symptoms analyses

Cough & Phlegm

- Nature of cough/ phlegm: dry cough/ sputum color?
- Course : acute/ subacute/ recurrent/ chronic?

Not all cough & phlegm are due to URIs!

- Emergencies : cyanosis/dyspnoea/drooling
- Serious: pneumonia, TB, Cancer and CHF
- Pitfalls: allergic rhinitis, asthma, COAD, bronchiectasis, GERD and drugs

Cough & Phlegm in URIs

- Acute irritating cough
- Scanty sputum, white/yellow
- Postnasal drip/ sorethroat
- Daytime & before or after sleeping
- GC good, no SOB/ chest sign
- Self-limiting, lasting 1-3 weeks

Runny Nose & Nasal Congestion

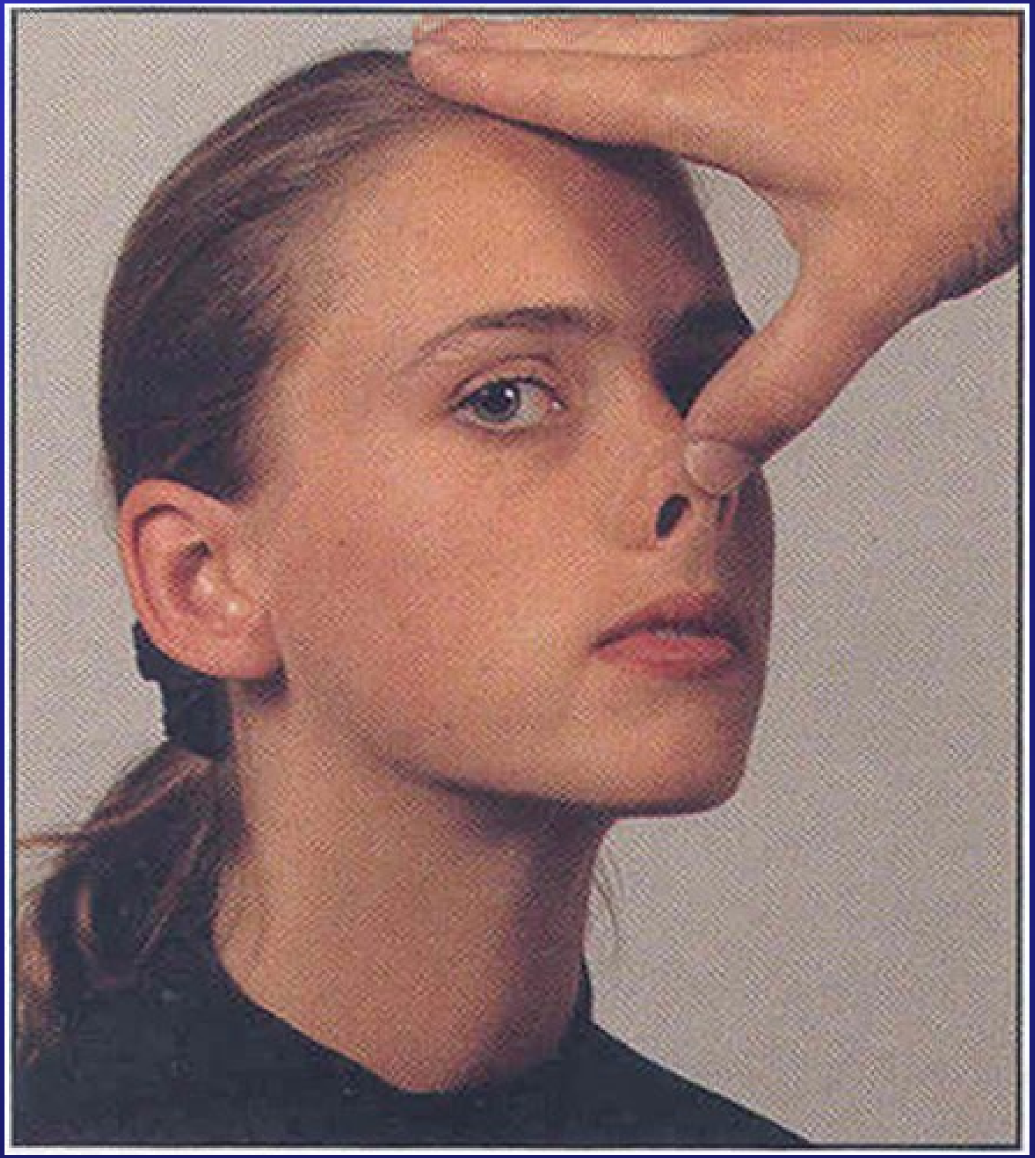
- Nature & course: yellow nasal discharge may, but does not necessarily, equal to bacterial infection

Not all runny nose are common colds!

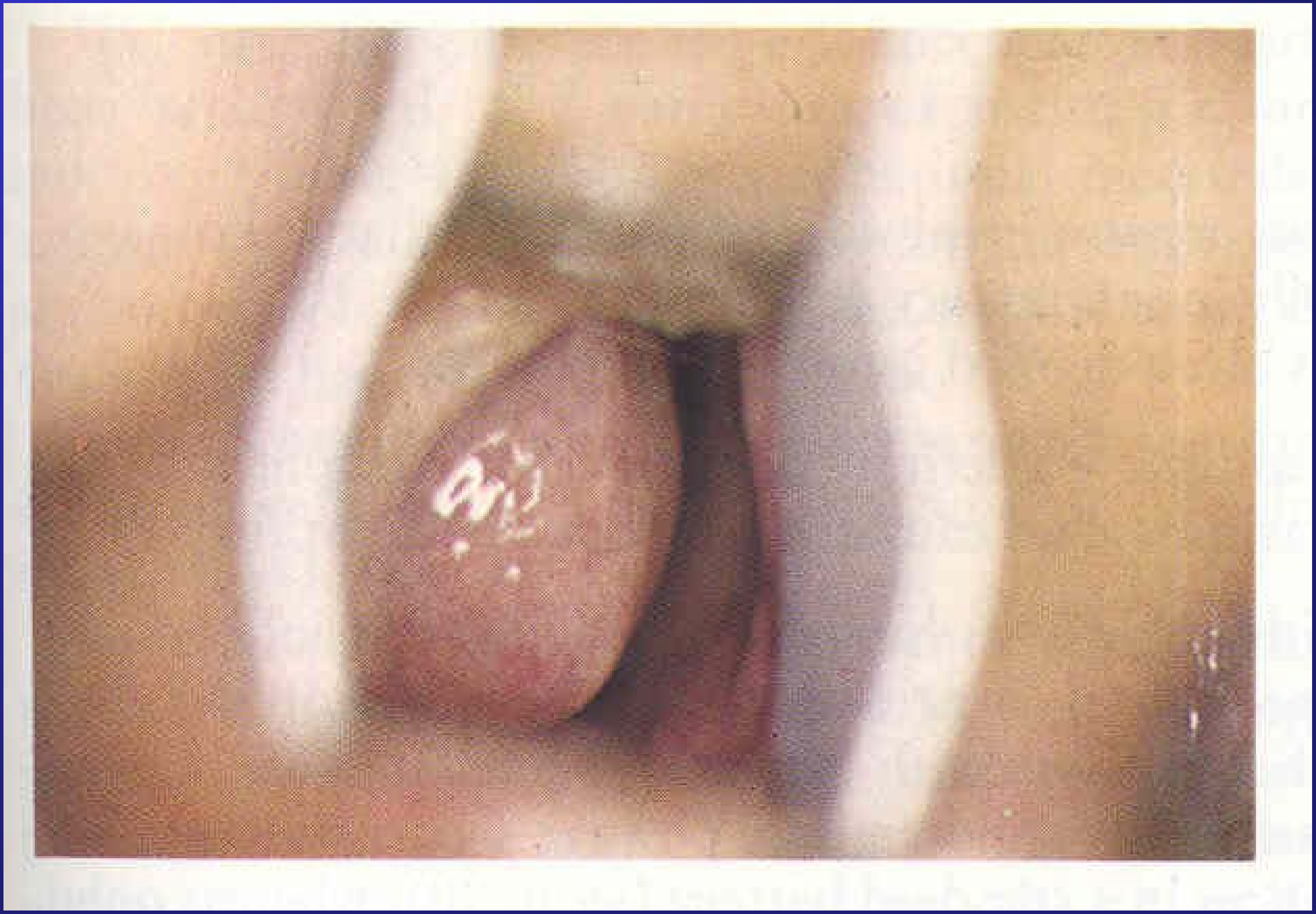
- Serious: NPC
- Pitfalls: allergy, polyps, foreign body & sinusitis
- A common cause of snoring & loss of smell
- Complications, e.g. otitis media, sinusitis

Nasal Symptoms in URIs

- Acute onset
- Copious clear watery discharge
- Sneezing++, relatively little itchiness
- Little diurnal variation
- Self-limiting, lasting 3-5 days



Reproduced from Epstein et al, Clinical Examination, 2nd Ed. Mosby 1997



Reproduced from Fobes CD, Jackson WF.
A Colour Atlas and Text of Clinical Medicine. Wolfe 1993.

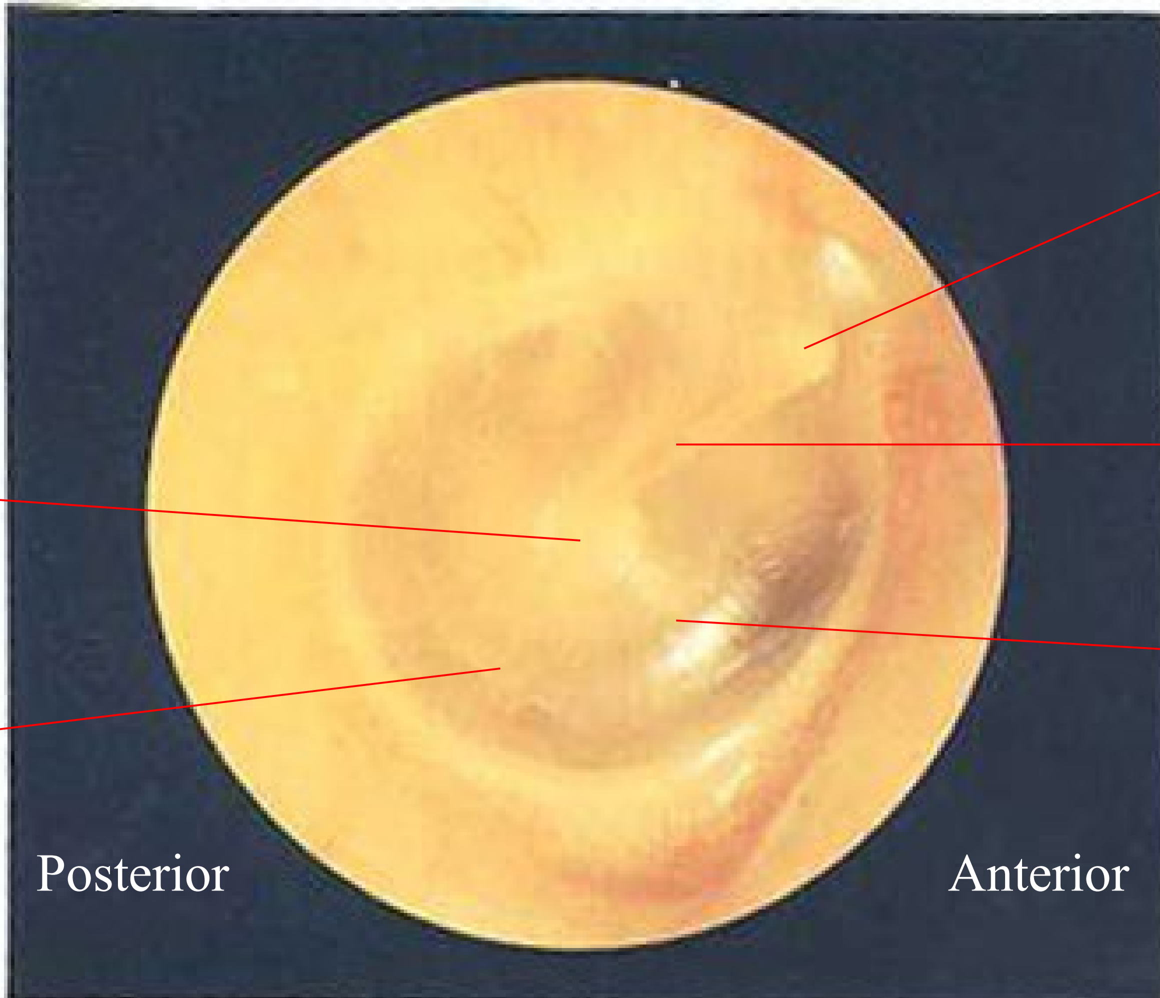


Fig. 4.29 Normal tympanic membrane. (Right ear)

Reproduced from Epstein et al, Clinical Examination, 2nd Ed. Mosby 1997

Sore Throat

- >90% acute sore throat due to viral URIs
- Emergency: acute epiglottitis (drooling of saliva)
- Serious : bacterial tonsillitis, abscess (quinsy), TB, lymphoma
- Pitfalls: postnasal drip, irritation, foreign body, infectious mononucleosis
- Enlarged tonsils are normal in children
- Exudates can mean viral or bacterial

CENTOR Criteria for sore throat

- Clinical use: Estimates probability that the pharyngitis is streptococcal, and suggests management course.
- **Four criteria** (1 point for each positive criterion):
  - History of fever
  - Tonsillar exudates
  - Tender anterior cervical adenopathy
  - Absence of cough
- Modified Centor Criteria (add the patient's age to the criteria):
  - Age 3-14 add 1 point
  - Age ≥45 subtract 1 point
- Scores range: -1 to 5

References:
1.Centor RM; Witherspoon JM; Dalton HP; Brody CE; Link K. The diagnosis of strep throat in adults in the emergency room. *Medical Decision Making*. 1981.1(3):239-246.
2.McIsaac WJ, White D, Tannenbaum D, Low DE. A clinical score to reduce unnecessary antibiotic use in patients with sore throat. *CMAJ*. 1998. Jan 13;158(1):75-83.

CENTOR Criteria for sore throat

A contemporary suggested guideline for management:

- -1, 0 or 1 points: No antibiotic or throat culture necessary.
- 2 or 3 points: Consider rapid strep testing and/or culture and treat with an antibiotic if result is positive.
- 4 or 5 points: Consider rapid strep testing and/or culture and treat with an antibiotic if result is positive. (Traditionally in McIsaac's paper in 1998 it was suggested to treat empirically with an antibiotic on clinical grounds.)
  - Remarks: The 2012 guidelines from the IDSA (Infectious Disease Society of America) no longer recommend empiric treatment for strep based on symptomatology alone; they recommend testing patients that are at higher risk for strep pharyngitis, but not giving antibiotics until a rapid test is positive or a throat culture is positive. The overall aim is to avoid inappropriate use of antibiotics.

Reference:

1. https://en.Wikipedia.org/wiki/Centor_criteria. Accessed date: 29 Nov 2022.
2. McIsaac WJ, White D, Tannenbaum D, Low DE. A clinical score to reduce unnecessary antibiotic use in patients with sore throat (1998). *CMAJ*. Jan 13;158(1):75-83.
3. Shulman ST, Bisno AL, Clegg HW, Gerber MA, Kaplan EL, Lee G, Martin JM, Van Beneden C. Clinical Practice Guideline for the Diagnosis and Management of Group A Streptococcal Pharyngitis:2012 Update by the Infectious Diseases Society of America. *Clinical Infectious Diseases*, Vol. 55(10) :e86-e102.

Hoarseness

- Indicates pathology in the larynx
- Most likely viral URI if acute
- DDx: vocal cord polyps/nodules
- Serious: acute epiglottitis, croup, carcinoma of larynx
- Pitfalls: sputum, laryngeal injury/compression, trauma from intubation

How to prevent URIs?

Transmission of the Common Cold

(Lorber B. *JGIM* 1996; 11:229-236)

- Transmission by contacts & droplets
- Maximum viral shedding on D2 & D3
- Virus found in 40% hand sample
- Natural infectivity rate 38-88%
- Median incubation 3 d (1-10)
- 70-90% infected are symptomatic
- Diagnosis is clinical; viral culture possible; serology useless

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Natural course of simple viral URTI

- Cough: 2-20 days
- Headache: 1-14 days
- Hoarseness: 2.5 -20 days
- Muscle ache: 2-14 days
- Runny nose: 2-14 days
- sorethroat: 2-14.3 days

Reference:
Wong W., Lam C. L. K., Fong D. Y. T. Treatment effectiveness of two Chinese herbal medicine formulae in upper respiratory tract infections – a randomized double-blind placebo-controlledtrial. *Family Practice* 2012: 29: 643-652.

Investigation

- Any investigation needed?
- 1. Chest X-ray?
- 2. Sputum / throat swab for culture?
- 3. Nasal Aspirate for culture?
- 4. Blood test?
- 5. Others?

Investigation

- Common cold?
- Influenza?
- Pharyngitis?

Learning Objectives

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Management of simple viral URI

- Body's immune system is most effective
- Mean resolution 2-3 days, 75th percentile resolution 7 days (cough up to 10 days)
- Stress reduction and rest are important
- Curative treatment is usually not needed
- Symptomatic treatment
  - Not a pill for every illness/symptom
  - evidence-based
  - benefit vs. harm
- Infection control -wash hands, wear masks

Symptomatic Treatments of URTIs

(Cochrane Library Systematic Review 2004)

- **Systemic symptoms/ sore throat**
  - Paracetamol is proven to be effective & safe
  - NSAID (ibuprofen) more effective but more side-effects
  - Aspirin is contraindicated in children with influenza (Reye's Syndrome)
  - No large-scale quality trials on lozenges
  - Steroids do more harm than good

Symptomatic Treatments of URIs

- **Cough and phlegm**
  - NO good evidence for the effectiveness of over-the-counter cough medicines[1]

Hence it is more important to:

- **Reassure patients on the expected course of cough in simple URI, eg. cough may persist up to 3 weeks**

Reference: Smith SM, Schroeder K, Fahey T. Over-the-counter (OTC) medications for acute cough in children and adults in community settings. Cochrane Database of Systematic Reviews 2014, Issue 11. Art. No.: CD001831. DOI: 10.1002/14651858.CD001831.pub5. Accessed 03 November 2023.

Symptomatic Treatments of URIs
(Schroeder K et al, *BMJ* 2002; 324:329-331)

- **Cough and phlegm**
  - **FDA:** no cough and cold product that contains a decongestant or antihistamine for kids <2y.o "because serious and possibly life-threatening side effects could occur."
  - **Health Canada/NHS:** no OTC cough and cold medicine to kids < 6

References:
Smith SM, Schroeder K, Fahey T. Over-the-counter (OTC) medications for acute cough in children and adults in community settings. Cochrane Database of Systematic Reviews 2014, Issue 11. Art. No.: CD001831. DOI: 10.1002/14651858.CD001831.pub5. Accessed 03 November 2023.

https://www.canada.ca/en/health-canada/services/drugs-medical-devices/concerns-about-children-s-medication.html

What is in Cough Mixture?

May contain (or in a combination of):

- Cough suppressant
- Expectorant
- Antihistamine
- Decongestant
- Mucolytic, e.g bromhexine, acetylcysteine
- Herb, e.g. Echinacea, squill

Cough Suppressants

(Simasek M et al. *Am Fam Phy* 2007; 75:515-520.)

- **Opioid (Not recommended)**
  - Codeine, Pholcodine, Dextromethorphan
  - Uncertain effectiveness for URTIs
  - Not recommended in children
  - Side effects: sedation, GI disturbances, etc.
- **Non-opioid (Sedating antihistamines)**
  - Diphenhydramine (Benadryl Expectorant), dexbrompheniramine
  - Inconsistent results from RCT
  - Side effects: blurred vision, dry mouth, urinary retention, etc.

Cough Expectorants

- Thin the mucus, making it easier to be coughed up.
- **Common preparations**
  - Ammonium chloride, e.g. MES
  - Ipecacuanha, e.g. MES
  - Guaifenesin, e.g. Robitussin
  - Squill, e.g. Cocillana compound syrup
- Effectiveness not proven by RCT, except some evidence on guaifenesin (Arroll B. *Resp Med* 2005; 99: 1477-1484.)
- High dose may cause nausea & vomiting

Other drugs for cough?

- Mucolytics: eg. Bromhexine
  - loosens and thins bronchial secretions by reducing surface tension and viscosity of mucus
  - Side effects: dizziness, headache, GI disturbance, etc.
- Inhaled or oral beta agonists
  - May be effective for prolonged cough in URI, esp. in patients with bronchial hypersensitivity.
  - Side effects: palpitations, tremor, etc.
  - Seldom necessary

Treatments for Nasal symptoms

- **Antihistamines**
  - Non-selective e.g. chlorpheniramine may reduces sneezing & rhinorrhoea
  - Non-sedating selective antihistamines are much less effective
- **Nasal decongestants**, e.g. pseudoephedrine, phenylpropanolamine (PPA)
  - May provide transient relief of nasal obstruction
  - Side effects are common and can be serious, caution in hypertensive patients

NO proven efficacy in children and adults for URI.

FDA warning: Avoid in children < 2 y.o.

https://www.fda.gov/drugs/resourcesforyou/specialfeatures/ucm263948.htm

Treatments for Nasal symptoms

(Simasek M et al. *Am Fam Phy* 2007; 75:515-520.)

- **Topical Ipratropium**
  - Topical ipratropium (Atrovent) is a treatment option for nasal congestion in children older than six years and in adults, although it is expensive.

Heated humidified air
  - Conflicting results
  - Yet, benign and possibly beneficial

Novel Treatments of Coryza

(Cochrane Library Systematic Review, 2004)

- **Vitamin C 1-3 g**
  - no benefit if taken at onset of illness
- **Zinc (>75mg/day)[1]**
  - May shorten illness by 1-3d in healthy people
  - Need high dose >75mg/day
  - Start within 24 hours of onset of symptoms and take throughout
  - Insufficient data regarding prophylaxis treatment

[1]Singh M, Das RR. Zinc for the common cold. Cochrane Database of Systematic Reviews 2013, Issue 6. Art. No.: CD001364. DOI: 10.1002/14651858.CD001364.pub4. Accessed 05 November 2023.

Combination Preparations for URTIs

(Sung V et al. *Australian Prescriber 2009; 32:122-4)*

- Antitussives, antihistamines, expectorants and decongestants
- Risk of overdosing in children and associate with sudden infant deaths.

Shotgun therapy* increases side-effects

- Beware of aspirin (salicylate) & Phenylpropanolamine (PPA)
- Antihistamine overdose from cold medicine plus cough mixture

*any treatment that has a wide range of effects and is expected to correct an abnormal condition even though the particular cause is unknown.

Should antibiotics or steroids be prescribed?

- Antibiotics and Steroids are not doing any good for viral URTI.
- Side-effects of drugs
- Antibiotics resistance

Which of the following is proven to be more effective in the prevention of the common cold?

1. **Daily Vitamin C**
2. **Daily 8 hours sleep**

SLEEP MATTERS!

- Participants with <7 hrs of sleep were 2.94 times more likely to develop a cold than those with >/=8 hrs of sleep.
- Participants with <92% sleep efficiency were 5.50 times more likely to develop a cold than those with >/=98% efficiency.
- These are independent of their pre-challenge virus-specific Ab titers, demographics, season of the year, body mass, socioeconomic status, psychological variables, health practices, or the percentage of days feeling rested.

Reference:
Cohen S., Doyle W. J., Alper C. M., Deverts D. J., Turner R. B. 2009. Sleep Habits and Susceptibility to the Common Cold. *Arch Intern Med*. Vol 169 (1):62-67

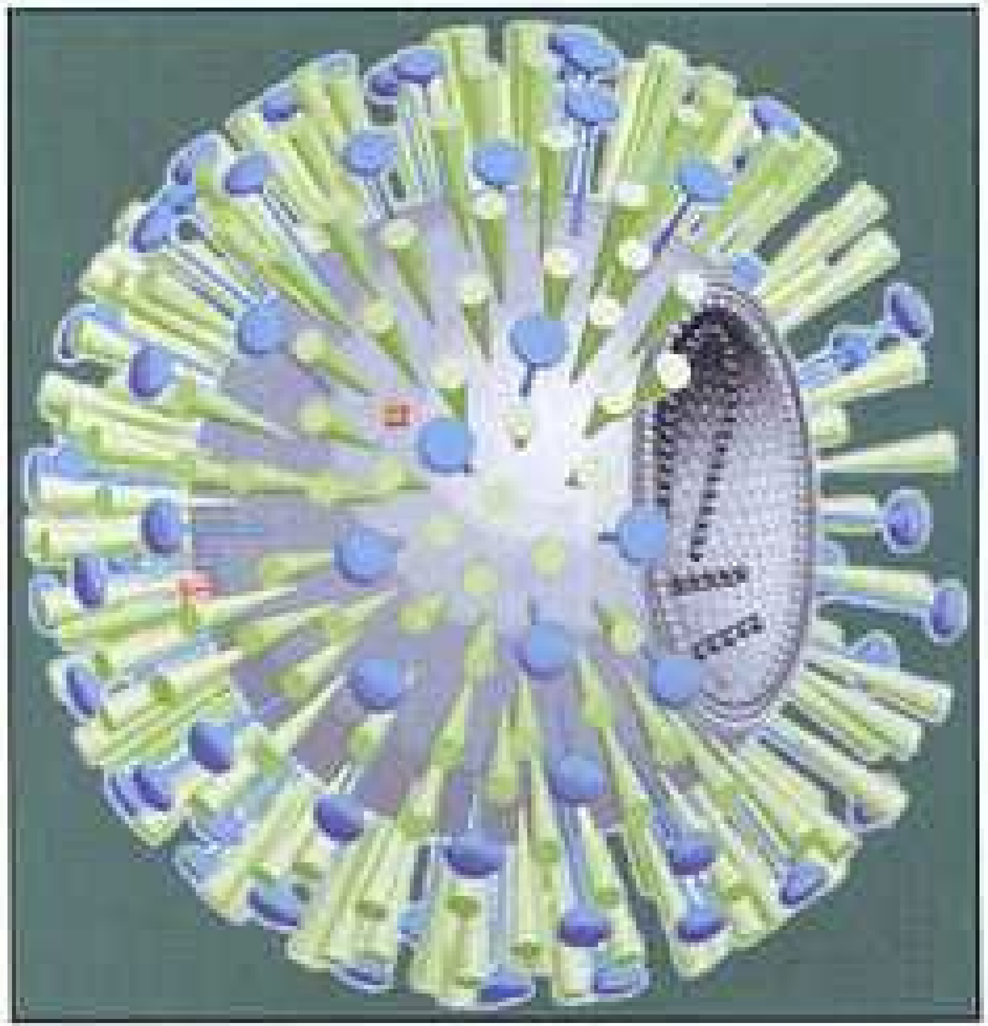

Figure 1. The Influenza virus.

Reproduced from *Aus Fam Phy* 2002; 31:145.

Influenza

- Three types of influenza viruses affecting human: A, B and C
- Criteria of ILI: fever ≥38 C + cough and /or sore throat
- Severe systemic upset
- Generalized myalgia
- Nasal symptoms mild
- Rapid test not more accurate than clinical Dx*
- Most cases are self-limiting, complications (mostly pneumonia) occur in 5-20%

* Reference: Stein J et al. *Ann Emerg Med 2005; 26: 412-419.*

Vaccination Against Influenza

- Effective in preventing illness, complications, hospitalization and death.

- The composition of the seasonal influenza vaccines in HK follows the recommendations by the World Health Organization and it varies every year.

- Inactivated influenza vaccines (most are given via IM)

  -For individuals aged 6 months or above

- Live attenuated influenza vaccine (intranasal)

  -Can consider for non-pregnant and non-immunocompromised people aged 2-49 years

  -children <5 with recurrent wheezing / patients with asthma may be at increased risk of wheezing after adminstration

- Recombinant influenza vaccine (registered in HK since 4/2021)

  -Created synthetically by recombinant technology and it does not require egg nor influenza virus during the production process

  - For individuals of 18 years or above

Reference:https://www.chp.gov.hk/files/pdf/recommendations_on_seasonal_influenza_vaccination_for_the_2023_24_season_in_hong_kong_19apr.pdf

Vaccination Against Influenza

- Efficacy: reduces risk of influenza by 40-60% when circularting viruses closely matched vaccine strains
- Recombinant vaccine MAY be more effective in older age group vs inactivated vaccine
- Should be given annually at the beginning of peak season (Oct-Nov), expected to be covering the winter and summer surge.
- Effective 2 weeks post-vaccination.

Reference:
https://www.chp.gov.hk/files/pdf/recommendations_on_seasonal_influenza_vaccination_for_the_2023_24_season_in_hong_kong_19apr.pdf

Priority Groups for flu vaccine

- Health Care workers
- Institutionalized persons
- Age 50 years or above
- People with chronic medical problems:
  - chronic lung/ cardiovascular (except uncomplicated HT per se)/ renal/metabolic diseases/obesity with BMI 30 or above/ immunocompromised/all children and adolescents on long term aspirin/people with chronic neurological condition that can compromise respiratory function or self-care ability or lead to increased risk of aspiration
- Children and adolescents (6 months to 18 years)
- Pregnant women
- Poultry workers/pig farmers/pig slaughtering industry personnel

Reference:
https://www.chp.gov.hk/files/pdf/recommendations_on_seasonal_influenza_vaccination_for_the_2023_24_season_in_hong_kong_19apr.pdf

Influenza Drug Treatments

(Centers for Disease Control and Prevention)

- **Oral Oseltamivir (Tamiflu®)**
- **Inhaled Zanamivir (Relenza®)**
- **Intravenous Peramivir (Rapivab®)**
- **Oral Baloxavir marboxil (Xofluza®)**

They can:
-shorten illness by about 1 day, if given within 48 hours
-reduce complications

Common side effects:
Oseltamivir: nausea and vomiting
Zanamivir: bronchospasm
Peramivir: diarrhoea

Summary

- URIs are common and highly infectious
- Not all URT symptoms are caused by infection
- Different URIs have different pathogenesis & natural courses
- Curative treatments are limited and usually unnecessary
- Benefit of symptomatic treatments needs to be balanced against harm
- Personal hygiene and influenza vaccine are very important in prevention
- Flu anti-viral drugs may be considered in patients with influenza depending on circumstances.

Further Reading

- Chris Del Mar. Acute sinusitis and sore throat in primary care. *Aust Prescr 2016;39:116-8.*
- Lam CLK. Rational prescribing for upper respiratory tract infections. *HK Pract* 2002; 24:298-304
- https://www.mdcalc.com/centor-score-modified-mcisaac-strep-pharyngitis
- Simasek M et al. Treatment of the common cold. *Am Fam Physician* 2007; 75: 515-520
- Aroll B. Non-antibiotic treatments for URTI (common cold). *Resp Med* 2005; 99: 1477-84
- Schroeder K et al. Systematic review of randomised controlled trials of over the counter cough medicines for acute cough in adults. *BMJ* 2002; 324: 329-31
- https://www.fda.gov/drugs/resourcesforyou/specialfeatures/ucm263948.ht m
- Sung V, Cranswick N. Cough and cold remedies for children. *Australian Prescriber 2009; 32:122-4*
- Centers for Disease Control and Prevention. What you should know about flu anti-viral drugs. https://www.cdc.gov/flu/treatment/whatyoushould.htm

Evaluation

TELL US WHAT YOU THINK

THANK YOU!

JC Whole Class Session

JC
WCS

Look for today's teacher
and the others you may have missed.
(HKU Portal login required)

Teacher: Dr Diana Wu

SAPIENTIA ET VIRTUS